

Report R18 — Archive or Encoder?

Paper 1’s first limitation was that every site-leakage number came from Phikon-v2 features. This re-runs the whole measurement over a second encoder that has never seen histology.

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Every number in this report is substituted from `evidence/results.json` by `make_report.py`. Nothing is typed by hand. This is the response to R19, where two manuscript numbers had no artefact behind them and the verification pass confirmed them against memory.

Summary

R12–R15 measured a large site signature in whole-slide features: subtype AUROC inflates by +0.1710 and the six methylation targets by +0.0849 in mean ρ when cross-validation folds share tissue-source sites instead of holding them out. Every one of those numbers came from Phikon-v2. The obvious objection is that pan-cancer histology pretraining might encode institutional stain and scanner character, in which case the finding is about Owkin’s model and not about federated pathology.

This report re-runs the identical measurement over `facebook/dinov2-large`: the same `Dinov2Model` class, the same 1024-dimensional width, the same 24 layers, the same DINOv2 self-supervision, the same CLS-token readout and the same ImageNet normalisation — trained on natural images rather than histology. The tile grids are byte-identical between the two feature sets, verified on slides spanning 887 to 28,666 tiles, so the encoder is the only thing that differs.

Verdict: SITE_SIGNATURE_IS_ARCHIVE_NOT_ENCODER. The site signature is a property of the archive, not of Phikon-v2. It reproduces under an encoder that has never seen a histology slide, so Paper 1’s claim is about federated pathology rather than about one vendor’s model.

1 The controls came first

A comparison across encoders is worthless if the harness is not the one that produced the published numbers, so the Phikon-v2 arms were re-run through the new harness and gated against R15 before the second encoder was allowed to count.

Control	Recomputed	Published	
subtype AUROC, site-grouped	0.7992	0.799	PASS
subtype AUROC, random folds	0.9703	0.9703	PASS
KEAP1 AUROC, site-grouped	0.6642	0.664	PASS
mean methylation inflation	0.0849	0.0849	PASS

All six grouped and all six random methylation correlations came back bit-identical to the stored values, which is what a deterministic seed should give and is worth stating because it means the comparison below is not absorbing harness noise.

2 Results

2.1 Subtype

Encoder	Site-grouped	Random	Inflation	Relative
owkin/phikon-v2	0.7992	0.9703	+0.1710	+0.8518
facebook/dinov2-large	0.7356	0.9390	+0.2034	+0.7694
KEAP1, owkin/phikon-v2	0.6642	0.6660	+0.0018	+0.0053
KEAP1, facebook/dinov2-large	0.5595	0.5797	+0.0202	+0.0458

Why the relative column carries the verdict. facebook/dinov2-large is not trained on histology, so its absolute performance is expected to be lower, and a smaller raw inflation could then reflect less available signal rather than less leakage. Relative inflation is (random – grouped)/(1 – grouped): the fraction of the remaining distance to a perfect score that fold assignment alone recovers. That is the comparison that survives one encoder being weaker, and it was pre-declared as the deciding quantity in the config before either dinov2 arm was read.

2.2 Methylation, six fixed targets

Target	owkin/phikon-v2			facebook/dinov2-large		
	grouped	random	infl.	grouped	random	infl.
KEAP1 signature	0.3172	0.3927	+0.0755	0.2139	0.1982	-0.0158
global	0.2897	0.3680	+0.0783	0.1748	0.1603	-0.0145
CpG island	0.2182	0.3358	+0.1176	0.0813	0.1231	+0.0418
open sea	0.2868	0.3478	+0.0610	0.1720	0.1723	+0.0003
TSS200	0.3029	0.4212	+0.1184	0.1855	0.1903	+0.0048
gene body	0.2988	0.3577	+0.0589	0.1673	0.1529	-0.0144
mean	0.2856	0.3705	+0.0849	0.1658	0.1662	+0.0004

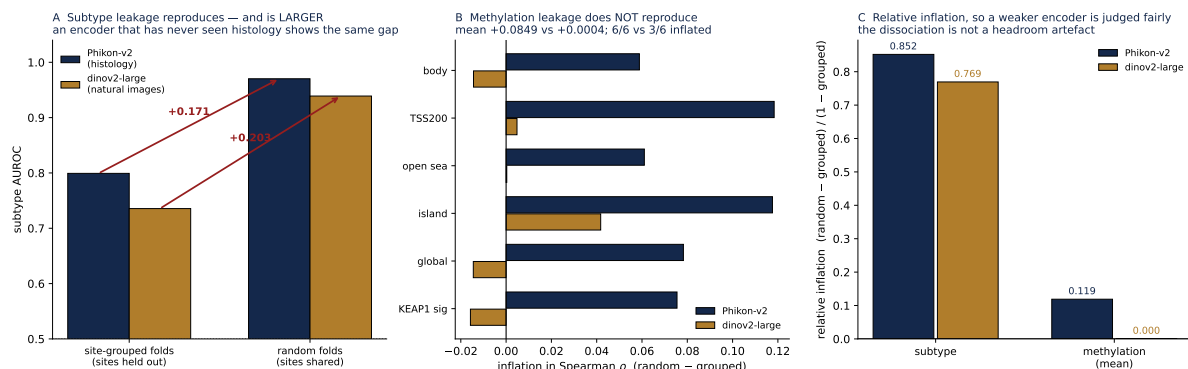
Targets inflated in the leakage direction: 6 of 6 for owkin/phikon-v2, 3 of 6 for facebook/dinov2-large.

Secondary verdict: METHYLATION_LEAKAGE_DID_NOT_REPRODUCE. The methylation leakage does NOT reproduce under this encoder, and the dissociation is the most informative thing in the report: subtype leakage is a property of the archive while methylation leakage is not universal. Paper 1 must split the claim rather than stating a single site-leakage magnitude. **Corrected 2026-08-17 by R21.** This report originally went further and called methylation leakage *a property of Phikon-v2 features*. That attribution is wrong. R21 added three pathology encoders – UNI, H-optimus-0 and Virchow2 – and every one inflates all six methylation targets (mean +0.0173 to +0.0849), while only the natural-image encoder does not (+0.0004, 3/6). The dissociation is by TRAINING CORPUS, not by vendor. The limitation that would have caught this – *two encoders is not a survey* – was stated in this report’s own limitations section and the inference was drawn anyway.

Relative inflation makes the dissociation harder to dismiss as a headroom effect. For facebook/dinov2-large the mean methylation inflation is +0.0004 in ρ against +0.0849 for owkin/phikon-v2 — not merely proportionally smaller but approximately zero, with three of six targets moving in the *anti*-leakage direction. A weaker representation with less signal to inflate would show a reduced gap; it would not show a gap that vanishes while the same architecture’s subtype gap is the larger of the two.

A reading, offered as a hypothesis and not a result. Phikon-v2 is trained on histology and encodes fine-grained stain and preparation character; methylation arrays for a given patient were run at the same institution that cut the slide, so batch structure in the assay and batch

structure in the image share an index. **facebook/dinov2-large**, trained on natural images, captures coarser morphology — enough for subtype at 0.7356 under site-disjoint folds, but its methylation signal is both weaker (0.1658 against 0.2856) and apparently not site-structured. Nothing here tests that mechanism.



A Subtype, both fold regimes, both encoders. **B** Per-target methylation inflation. **C** The dissociation on relative inflation, so the weaker encoder is judged on fraction-of-headroom rather than raw gap.

3 Limitations

Two encoders is not a survey. This distinguishes "a property of Phikon-v2" from "a property of the archive". It does not establish how the signature behaves across the pathology foundation-model field. UNI, H-optimus-0 and Virchow2 are gated behind licence acceptance that was not ours to give.

The contrast is corpus, and also patch size. **facebook/dinov2-large** uses patch 14 against Phikon-v2's 16, so at 224 px it sees 256 tokens rather than 196. Architecture class, width, depth, objective, readout and normalisation are matched; patch size is not, and it cannot be without retraining one of them.

Absolute performance is not the point and should not be quoted as a comparison of encoders. A natural-image model doing worse at lung subtype than a histology model is expected and uninteresting. The claim concerns the *gap* between fold regimes within each encoder.

One cohort. TCGA lung, the same 760 patients, the same 67 sites, the same slides.

Reproducing this report

`evidence/m16_encoder_compare_analysis.py` (the analysis), `evidence/mil_encoder_compare.py` (the eight arms), `evidence/encode_dinov2.py` (the encoder, pinned revision), `evidence/config.yaml` (pre-declared rules, SHA-256 143b3b9d0a92fabbb...), `evidence/gates.json`, `evidence/results.json`, `evidence/arms/` (the eight raw arm JSONs). This document: `python3 make_report.py`.